

YATIN KUMAR

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RESEARCH PROFILE

Ph.D. Research Scholar in Mathematics and Scientific Computing at the National Institute of Technology Hamirpur, India. My research focuses on scientific machine learning and numerical methods for fractional, nonlinear, and nonlocal differential equations. I develop neural-network-based solvers using physics-informed neural networks, Kolmogorov-Arnold networks, wavelet neural networks, interpolation-based methods, and operator learning.

RESEARCH GOAL

To develop reliable, interpretable, and computationally efficient scientific machine learning methods for fractional and high-order differential equations, supported by rigorous numerical analysis. The goal is to enable more accurate and efficient modelling of complex physical systems in heat transfer, fluid flow, phase-field dynamics, and pollutant transport, contributing to engineering design, environmental monitoring, and sustainable technological development.

RESEARCH INTERESTS

- Deep Learning for Scientific Computing
- Kolmogorov-Arnold Networks
- Neural Operators and Operator Learning
- Physics-Informed Neural Networks
- Computational Mathematics
- Fractional Differential Equations

EDUCATION

2023-Present	Ph.D. in Mathematics and Scientific Computing National Institute of Technology Hamirpur, Himachal Pradesh, India. Title: Deep Learning Algorithms for Approximation of Fractional Differential Equations. Expected completion: April, 2027.
2019-2021	M.Sc. in Mathematics Hemvati Nandan Bahuguna Garhwal University, Srinagar Garhwal, Uttarakhand, India.
2016-2019	B.Sc. in Physics, Chemistry, and Mathematics D.A.V. (P.G.) College, Dehradun, Uttarakhand, India.

DOCTORAL ADVISOR

Dr. Ramesh Kumar Vats
Associate Professor, Department of Mathematics and Scientific Computing
National Institute of Technology Hamirpur, Himachal Pradesh, India

JOURNAL PUBLICATIONS

- **Yatin Kumar** and Ramesh Kumar Vats, “A Deep Kolmogorov-Arnold Network Framework for Solving Time-Fractional Partial Differential Equations,” *Applied Soft Computing*, Article 114379, 2025. DOI: 10.1016/j.asoc.2025.114379
- **Yatin Kumar**, Ramesh Kumar Vats, Ankit Kumar Nain, and Sanjeev Yadav, “Robust Deep Learning Framework Using Hermite Interpolation for Time-Dependent Caputo Fractional Differential Equations,” *Mathematical Methods in the Applied Sciences*, vol. 49, no. 7, pp. 6398-6427, 2026. DOI: 10.1002/mma.70395
- **Yatin Kumar**, Ramesh Kumar Vats, Ankit Kumar Nain, and Sanjeev Yadav, “Deep Wavelet Neural Network with Lateral Connections for Solving Fractional Differential Equations Arising in Oil Pollution Dynamics,” *International Journal of Computer Mathematics*, pp. 1-43, 2026. DOI: 10.1080/00207160.2026.2663999

BOOK CHAPTER

- **Yatin Kumar** and Ramesh Kumar Vats, “Novel Deep Neural Network Framework for Nonlinear Time Fractional Differential Equations,” in *Artificial Intelligence Modeling for Dynamical Problems*, Publisher: Elsevier, ISBN: 9780443491344, Status: Accepted.

CONFERENCE PROCEEDING

- **Yatin Kumar**, Ramesh Kumar Vats, and Sanjeev Yadav, “Numerical simulation of fractional-order Bagley-Torvik equation using Pell polynomial neural network,” in *Proceedings of the Nonlinear Applied Analysis and Optimization*, Springer Nature, 2026. Status: Accepted.

CONFERENCE PRESENTATIONS

- 2026** “Advanced Deep Learning-Based Neural Solver for Time-Fractional Phase-Field Models” Presented at the *Scientific Computing and Differential Equations Conference (SciCADE 2026)*, University of Edinburgh, Edinburgh, United Kingdom, 29 June to 3 July 2026.
- 2026** “A Unified Deep Learning Framework for Solving Generalized-Order Differential Equations” Presented at the First International Conference on Mathematics Optimization Theory and Application (ICMOTA 2026), IIT BHU, India, 14 to 16 March 2026.
- 2024** “Numerical Simulation of Fractional-Order Bagley-Torvik Equation Using Pell Polynomial Neural Network” Presented at the 4th International Conference on Nonlinear Applied Analysis and Optimization (ICNAAO 2024), National Institute of Technology Hamirpur, India, 17 to 19 October 2024.

WORKSHOPS AND PROFESSIONAL DEVELOPMENT

- Attended International Faculty Development Programme on “Recent Trends in Differential Equations and Numerical Computing,” National Institute of Technology Calicut, India, 20 to 24 June 2025.

ONGOING RESEARCH

- Dynamic Loss Balancing Self-Adaptive Neural Solver for Nonlinear Time-Fractional Korteweg-de Vries Equations with Rigorous Error Analysis (**Under Review**).
- Learnable Deep Neural Solver for Numerical Simulation of Nonlocal Time-Fractional Phase-Field Models. (**Under Review**).
- GeoDiv-KAN Solver for Time-Fractional Navier–Stokes Equations on Complex Domains to Realistic CFD Reconstruction (**Under Review**).

AWARDS AND FELLOWSHIPS

- Qualified the Graduate Aptitude Test in Engineering (GATE) 2023, a national-level competitive examination in India.
- Awarded the Junior Research Fellowship (JRF) and subsequently the Senior Research Fellowship (SRF) by the Ministry of Education, Government of India, in support of doctoral research at the National Institute of Technology Hamirpur.
- Received international travel support from the Anusandhan National Research Foundation (ANRF), Government of India, to attend and present research at *Scientific Computing and Differential Equations (SciCADE 2026)*, University of Edinburgh, United Kingdom.

TECHNICAL SKILLS

Programming and numerical computing

Python, MATLAB, Maple

Scientific machine learning

Physics-informed neural networks, Kolmogorov-Arnold networks, Wavelet neural networks, Deep operator learning.

Numerical analysis

Fractional-derivative approximation, nonlinear differential equations, error analysis, numerical simulation

Documentation and productivity

LaTeX, Microsoft Word, Excel, PowerPoint

LANGUAGES

English - Professional proficiency | **Hindi** - Native proficiency